

Farid Zahran

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PROFILE

Computer Science student and full-stack software engineer transitioning into AI/ML engineering, with hands-on experience building and deploying production-grade web applications end-to-end. Actively building applied ML/AI projects to strengthen practical knowledge and experience in the field, combining strong software engineering fundamentals with practical machine learning implementation to ship reliable, deployable AI products.

EDUCATION

Mikroskil University

Bachelor's Degree in Computer Science

Medan, Indonesia

Sep. 2022 – Present

Relevant Coursework: Artificial Intelligence, Machine Learning, Computer Vision, Data Engineering, Statistics for Computing, Database Modeling and Implementation, Cloud Computing, Development and Operations, Back-End & Front-end Web Development, Back-End & Front-end Mobile Development

EXPERIENCE

Chief Technology Officer & Software Engineer

NextLevel Academy

Nov. 2024 – Present

Medan, Indonesia

- Co-founded and lead technical direction of a gamified e-learning startup, architecting and building a multi-role platform (admin, students, interns, mentors) from scratch to production — integrating payment gateway transactions, a custom EXP/leveling gamification engine, and an end-to-end internship management module — live at nextlevelacademy.id.
- Implemented role-based access control across 4 distinct user roles and secure invitation-link admin provisioning, currently operating in production for active internship management workflows.
- Optimized platform performance to achieve a 100 Lighthouse Performance score and sub-1s page load time, while maintaining 99.6% uptime in production.

Fullstack Developer (Internship)

PT. Kodinglab Integrasi Indonesia

Nov. 2024 – Jan. 2025

Medan, Indonesia

- Led continuous requirements-gathering through stakeholder interviews with university Student Affairs and the Student Executive Board (BEM) to define the goals, scope, and core features of a student organization (UKM) monitoring and management system, built from scratch in an industry-simulated development workflow.
- Designed the complete system architecture and documentation, including user activity flow diagrams, UI/UX wireframes in Figma, and database schema (ERD and attribute tables), presenting progress to mentors on a scheduled review cadence.

PROJECTS

NextLevel Academy | *Next.js, Supabase, Prisma, Better Auth, Midtrans, Bunny.net, Redis*

Nov. 2024 – Present

- Engineered a secure payment pipeline with Midtrans, using PostgreSQL advisory locks to prevent race conditions on concurrent orders and SHA-512 signature verification with idempotent webhook handling to ensure reliable, tamper-proof transaction fulfillment.
- Built a protected video streaming system on Bunny.net using HLS adaptive streaming and time-limited signed embed URLs, gated by server-side enrollment checks to prevent unauthorized content access and downloads.
- Designed a custom authentication layer with Edge/Node-split role-based access control, Redis-backed persistent rate limiting, and hardened account security including anti-enumeration protection and forced session invalidation on admin actions.
- Modeled a 41-entity PostgreSQL schema (Prisma) spanning course management, gamification, and internship modules, validated through 88 black-box test scenarios across 5 user roles with a 100% pass rate confirmed via user acceptance testing.

Contexture | *Next.js, FastAPI, Qdrant, Gemini, Groq*

Aug. 2026

- Built a RAG-based technical support chatbot for industrial equipment datasheets (motors, drives, sensors, PLCs), enabling engineers to get instant, accurate answers instead of manually searching dense multi-page PDF specification sheets.
- Designed a confidence-based guardrail so the system refuses to answer when retrieval confidence is low, reducing the risk of AI hallucination on safety-critical technical specifications.

- Implemented a resilient dual-provider LLM setup with automatic failover, plus real-time streaming answers with clickable source citations so users can verify every response against the original document.
- Deployed to production with a decoupled architecture (Vercel for frontend, Hugging Face Spaces for backend), live and fully functional at contexure-ai.vercel.app.

Industrial Predictive Maintenance | *Python, XGBoost, Scikit-learn, Streamlit, Plotly*

July. 2026

- Built a dual-layer machine failure detection system for industrial equipment sensor data (temperature, rotational speed, torque, tool wear), combining a supervised classifier for known failure patterns with an unsupervised anomaly detector to catch previously-unseen failure signatures.
- Tackled severe class imbalance (failures $\approx 3.4\%$ of data) using cost-sensitive training, achieving 78% recall and 98% accuracy on failure detection — prioritizing catching real failures over raw accuracy, since a missed failure is far costlier than a false alarm in industrial settings.
- Deployed a live interactive dashboard simulating real-time sensor monitoring with color-coded risk alerts (healthy / warning / critical), live at hoseki7-predictive-maintenance.streamlit.app.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, PHP

AI/ML & Data: Scikit-learn, XGBoost, Pandas, NumPy, RAG Systems, Vector Search, LLM Integration (Gemini, Groq, etc)

Frameworks: Next.js, React, FastAPI, Node.js, Express.js, Laravel, Tailwind CSS

Databases: PostgreSQL, MySQL, MongoDB, Prisma, Qdrant (Vector DB), Supabase

Tools: Git, Docker, Redis, Figma, Google Colab, VS Code, Claude Code, Antigravity